Early rows of the reflected triangles, firing classes, and the five-state near misses

Research notes accompanying the note
Two necessary conditions on minimal-time solutions of the firing squad problem

October 2026

Notation and numbering of results refer to the note: the half-line C_∞, the line C_n, the reflected triangle R_n in the relative coordinates $\tau = t-(n-1)$, $\kappa = n-i$, so that $R_n = \{(\tau,\kappa) : 0 \le \kappa \le \tau \le n-1\}$, the input word

$$J_n(\kappa) = \big(\Phi_{n-3+\kappa}(2\kappa-2),\ \Phi_{n-2+\kappa}(2\kappa-1)\big), \qquad 0 \le \kappa \le n-1,$$

the depth rows $\Phi_t(j) = C_\infty(t, t+1-j)$ with the recurrence

$$\Phi_{t+1}(j) = \delta(\Phi_t(j), \Phi_t(j-1), \Phi_t(j-2)) \qquad (0 \le j \le t+1), \tag{1}$$

where $\Phi_t(-1) = \Phi_t(-2) = \mathsf{L}$ and $\Phi_t(t+1) = *$ (equation (3) of the note), and the partial map Ψ of the determinacy lemma (Lemma 3.1 of the note). We write $R_n(\tau,\kappa) := C_n(n-1+\tau, n-\kappa)$. All statements are proved below; the computer-assisted parts are simulations of explicitly given rule tables and use no SAT solver. They are reproduced by the programs `asymp.py` and `failall.py` in the directory `fssp5/research` of the repository.

1 Early rows

For $T \ge 0$ let $E(T) := \{(\tau,\kappa) : 0 \le \kappa \le \tau \le T\}$. If $n \ge T+2$, then $E(T) \subseteq R_n$ and no cell of $E(T)$ lies in the firing row $\tau = n-1$ of R_n.

Lemma 1 (early cells)**.** *Let δ be a rule, $T \ge 0$, and $n, n' \ge T+2$ with $J_n(\kappa) = J_{n'}(\kappa)$ for $0 \le \kappa \le T+1$. Then for every $(\tau,\kappa) \in E(T)$, $\Psi(J_n, n; \tau, \kappa)$ is defined if and only if $\Psi(J_{n'}, n'; \tau, \kappa)$ is defined, and then the two values are equal. In particular, the line of length n has a cell in state F at a relative time $\tau \le T$ if and only if the line of length n' has one at the same relative position.*

Proof. By the definition of Ψ in the proof of Lemma 3.1 of the note,

$$\Psi(J, m; \tau, \kappa) = \delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1}),$$

where $x_{\kappa'}$ is the border $*$ if $\kappa' \in \{-1, m\}$, a component of $J(\kappa')$ if $\tau - 1 \in \{\kappa'-2, \kappa'-1\}$, and $\Psi(J, m; \tau-1, \kappa')$ otherwise (then $0 \le \kappa' \le \tau-1$). Unfolding this recursion from $(\tau,\kappa) \in E(T)$ visits only pairs (τ'', κ'') with $0 \le \kappa'' \le \tau'' \le T$ and reads the symbols $x_{\kappa'}$ with $\kappa' \le \kappa''+1 \le T+1$. The case $\kappa' = m$ would require $m \le T+1$, which is excluded for $m \in \{n, n'\}$; hence the left border is never read, and the input word is read only at positions $\kappa' \le T+1$, where J_n and $J_{n'}$ agree.

The two recursions are therefore identical step by step, including the points where an argument equal to F makes δ undefined. The last statement follows because the cells of the line of length n outside R_n coincide with cells of C_∞ (Lemma 2.2 of the note) and are never F before the line has fired, while the cells of R_n at the relative times $\tau \le T$ are exactly the cells of $E(T)$. □

Lemma 2 (periodicity certificate)**.** *Let δ be a rule whose half-line does not fire, $D \ge 0$, and $V_t := (\Phi_t(0), \dots, \Phi_t(D))$ for $t \ge D$. There is a map G_D, depending on δ and D only, with $V_{t+1} = G_D(V_t)$ for all $t \ge D$. Consequently, if $V_{t_0} = V_{t_0+p}$ for some $t_0 \ge D$ and $p \ge 1$, then $V_t = V_{t+p}$ for all $t \ge t_0$.*

Proof. For $t \ge D$ and $0 \le j \le D$ we have $j \le t$, so the three arguments in (1) are $\Phi_t(j)$, $\Phi_t(j-1)$, $\Phi_t(j-2)$ with indices at most D (and L for negative indices); none of them is the border symbol $\Phi_t(t+1) = *$. Hence $G_D(v)_j := \delta(v_j, v_{j-1}, v_{j-2})$, with $v_{-1} = v_{-2} = \mathsf{L}$, does it; it is well defined on the vectors that occur because the half-line does not fire. The second statement follows by induction on t. □

Corollary 3 (finitely many early-row problems)**.** *Let δ, $T \ge 0$, $D := 2T+1$, $t_0 \ge D$ and $p \ge 1$ satisfy $V_{t_0} = V_{t_0+p}$. Then $J_n(\kappa) = J_{n+p}(\kappa)$ for all $0 \le \kappa \le T+1$ and all $n \ge t_0 + 3$. Hence, for every $N_0 \ge T+2$, each line of length $n \ge N_0$ has the same cells at the relative times $\tau \le T$ (in the sense of Lemma 1) as one of the lines of the finitely many lengths $N_0 \le n' < \max(N_0, t_0+3) + p$.*

Proof. For $0 \le \kappa \le T+1$ the two components of $J_n(\kappa)$ are depth-row cells of depths $2\kappa-2, 2\kappa-1 \le 2T+1 = D$ at the times $n-3+\kappa$ and $n-2+\kappa$, which are at least $n-3 \ge t_0$ (for $\kappa = 0$ the lower component is L by convention, and the upper one is $C_\infty(n-2, n) = \mathsf{L}$ by Lemma 2.1 of the note). By Lemma 2 these cells are unchanged when n is replaced by $n+p$. If $n \ge \max(N_0, t_0+3) + p$, subtracting a suitable multiple of p gives n' in $[\max(N_0, t_0+3), \max(N_0, t_0+3)+p)$ with $J_{n'} = J_n$ on $[0, T+1]$, and Lemma 1 applies since $n, n' \ge N_0 \ge T+2$. □

Remark 4. For a minimal-time solution, every line of length $n \ge T+2$ has no F at the relative times $\tau \le T$. If the depth column V_t for $D = 2T+1$ is periodic from t_0 with period p, the early rows of R_n for $n \ge t_0 + 3$ depend only on $n \bmod p$: they are the first $T+1$ rows of finitely many *limit triangles*, which grow from the right border on the periodic background formed by the depth rows near the front. A minimal-time solution needs every limit triangle to stay free of F in all rows, just as the half-line itself never fires. The near misses of the note violate exactly this, as the next proposition shows.

Proposition 5. *None of the rules δ_{14}, δ_{14}^{b}, δ_{14}^{c}, δ_{14}^{d}, δ_{14}^{e} (files `results/delta14*.txt` of the repository) synchronizes a line of length $n \ge 15$ at time $2n-2$. Each of them therefore synchronizes exactly the lines of lengths $2, \dots, 14$ in minimal time.*

Proof (computer-assisted, by simulation). For each rule, Table 1 gives T and a repetition $V_{t_0} = V_{t_0+p}$ of the depth column for $D = 2T+4 \ge 2T+1$, found by simulating the half-line. Each rule's half-line uses only the transitions of its germ and does not fire on the computed cells, so Lemma 2 applies. By direct simulation:

(i) every line of length $15 \le n < T+3$ fails to synchronize at time $2n-2$ (early or incomplete firing);

(ii) for every n with $\max(15, T+3) \le n < t_0 + 3 + p$, the line of length n, or a line of a length n' in the same range with $J_{n'} = J_n$ on $[0, T+1]$, has a cell in state F at a relative time $\tau \le T$.

By Lemma 1 and Corollary 3 (with $N_0 = \max(15, T+3)$), every line of length $n \geq \max(15, T+3)$ has an F at a relative time $\tau \leq T < n-1$, that is, before time $2n-2$. The lines of lengths $2, \ldots, 14$ are synchronized by these rules (Proposition 5.2 of the note and its text). □

rule	T	t_0	p	lengths simulated	first F at relative time
δ_{14}	10	51	8	35 $(15, \ldots, 49)$	$7, \ldots, 10$
δ_{14}^{b}	14	66	3	39 $(15, \ldots, 53)$	$8, \ldots, 14$
δ_{14}^{c}	22	257	192	413 $(15, \ldots, 427)$	$7, \ldots, 22$
δ_{14}^{d}	12	95	48	116 $(15, \ldots, 130)$	$6, \ldots, 12$
δ_{14}^{e}	14	56	2	28 $(15, \ldots, 42)$	$9, \ldots, 14$

Table 1: Certificates for Proposition 5: the depth column of depths $0, \ldots, 2T+4$ repeats with period p from time t_0; one length per distinct signature (input cells at $\kappa \leq T+2$ and the return-chain values of C_∞, a slightly finer partition than needed) is simulated, together with every length $15 \leq n < T+3$. Program: `failall.py RULE T`.

The early-rows test Let g be a germ that determines C_∞ on the cells used below, and let the depth column of C_∞ for $D = 2T+1$ satisfy $V_{t_0} = V_{t_0+p}$. If a rule $\delta \supseteq g$ is a minimal-time solution, then for each of the finitely many lengths n of Corollary 3 no cell of R_n with $\tau \leq \min(T, n-2)$ is F, and $R_n(\tau, \tau) \neq C_\infty(n-1+\tau, n-\tau)$ for $\tau \leq \min(T, n-1)$ by Lemma 2.3 of the note. By Corollary 3, these conditions for the listed lengths imply the same conditions for every $n \geq N_0$: no further length adds a constraint on the early rows. Combined with the complete triangles of the lengths $2, \ldots, N_0 - 1$, they form a finite formula; if it has no model extending g, no minimal-time solution contains g. This test is the encoding `early:n:T` of `gfix.py`, with the lengths produced by `asymp.py`.

2 Early-row formulas

The early-rows test of Section 1 fixes a germ. The following finite formulas leave the whole rule free and combine complete lines with the early rows of longer lines.

Definition 6. Let $k \geq 3$, $T \geq 1$ and $N_1 \geq T+2$. A k-state rule δ satisfying the quiescence condition (1) of the note *satisfies* $\mathcal{G}_k(T, N_1)$ if

(G1) δ synchronizes the line of length n at time $2n-2$ for $2 \leq n \leq T+1$;

(G2) for $T+2 \leq n \leq N_1$, no cell of the line of length n is F at a time $t \leq n-1+T$;

(G3) for $T+2 \leq n \leq N_1$ and $1 \leq i \leq n$ with $t := 2n-1-i \leq n-1+T$, $C_n(t, i) \neq C_\infty(t, i)$;

(G4) $C_\infty(t, i) \neq$ F whenever $t + i \leq 2N_1$.

Lemma 7. *Every k-state minimal-time solution satisfies $\mathcal{G}_k(T, N_1)$ for all $T \geq 1$ and $N_1 \geq T+2$.*

Proof. (G1) holds by definition. For (G2), $n \geq T+2$ gives $2n-2 = (n-1)+(n-1) \geq (n-1)+(T+1) > n-1+T$, so the times $t \leq n-1+T$ precede the firing time $2n-2$ of the line of length n, at which the line is synchronized for the first time. (G3) is Lemma 2.3 of the note, which applies because the lines of lengths n and $n+1$ are synchronized at times $2n-2$ and $2n$. (G4) is equation (2) of the note. □

By (G1) and (G2), the cells of R_n that $\mathcal{G}_k(T, N_1)$ constrains are those of $E(T)$ for $n \geq T+2$ and all of R_n for $n \leq T+1$, which is why the lines $2, \ldots, T+1$ are complete and no shorter list of complete lines occurs in the definition. In the encoding of `gfix.py`, $\mathcal{G}_k(T, N_1)$ is `full:2-(T+1)` `early:(T+2)-`N_1`:T` in the free mode, with the half-line encoded cell by cell up to the anti-diagonal $2N_1$ (option `GFIX_MARGIN=2`, the default). The option `GFIX_SHALLOW` must not be used here. It encodes only the depth rows near the front for the long lines, and hence drops (G4) on the deeper cells of the half-line. The resulting relaxation for $T = 9$, $N_1 = 60$ has a model whose general fires at time 22.

Proposition 8 (computer-assisted; the unsatisfiable case is LRAT-certified)**.** *The formula $\mathcal{G}_4(6, 10)$ is unsatisfiable: no four-state rule satisfying (1) synchronizes the lines of lengths $2, \ldots, 7$ in minimal time and satisfies (G2)–(G4) for $n \in \{8, 9, 10\}$ and $T = 6$. Each of the following weakenings is satisfiable: $\mathcal{G}_4(5, 30)$; $\mathcal{G}_4(6, 9)$; and $\mathcal{G}_4(6, 10)$ without (G3).*

Proof. The program `gfix.py` encodes $\mathcal{G}_4(6, 10)$, with structural sharing (option `GFIX_SHARE=1`), as a CNF formula with 919 variables and 22,960 clauses; the SHA-256 digest of the CNF file is

`0e0ba286ec7b60d05f6edee470b5cbd4d6655cd184186e630a96bd4187932e03`.

CaDiCaL 3.0.1 refutes it in 41 s and writes an LRAT proof of 167 MB, which `lrat-check` verifies in 5 s (script `certify_lrat.sh`). For each weakening, CaDiCaL returns a model of the encoding, i.e. a partial rule, and the encoding is equivalent to the formula on the cells it contains. Without (G3) means option `GFIX_NORC=8,9,10`. □

Theorem 5.1 of the note refutes four states with the complete lines $2, \ldots, 9$, and the bound 9 is sharp for complete lines: 27 partial rules synchronize the lines $2, \ldots, 8$. Proposition 8 replaces the complete lines 8 and 9 by the first seven rows of the reflected triangles of the lines 8, 9, 10, together with their return chains.

The early rows only forbid firing cells. A requirement that some cell *does* fire is added by the right ends. For $C \geq T+2$, a rule satisfies $\mathcal{G}_k(T, N_1, C)$ if it satisfies $\mathcal{G}_k(T, N_1)$ and

(G5) for $T+2 \leq n \leq C$, writing $\Psi_n(\tau, \kappa) := \Psi(J_n, n; \tau, \kappa)$ for the determinacy map of Lemma 3.1 of the note: $\Psi_n(\tau, \kappa)$ is defined and different from F for all $(\tau, \kappa) \in R_n$ with $\tau + \kappa \leq n-1$ and $\tau < n-1$, $\Psi_n(n-1, 0) = \mathsf{F}$, and $\Psi_n(\kappa, \kappa) \neq C_\infty(n-1+\kappa, n-\kappa)$ for $2\kappa \leq n-1$.

The cells of (G5) are the backward dependency cone of the firing cell $(n-1, 0)$, and by Lemma 3.2 of the note the values Ψ_n there depend only on the input cells of that cone. So (G5) is a condition on the cone alone: a cell outside it may be F earlier, in which case the line of length n is no longer defined at time $2n-2$, while Ψ_n still is on the cone. In `gfix.py` it is the region `cone:`n, which encodes exactly these values. If no cell of the line of length n is F before time $2n-2$, then $\Psi_n = C_n$ on R_n by Lemma 3.1. In particular every minimal-time solution satisfies (G5), by the definition of synchronization and Lemma 2.3 of the note, and Lemma 7 extends to $\mathcal{G}_k(T, N_1, C)$. The conditions (G1)–(G4) need no such convention, because the cells they constrain form whole rows of each line.

Proposition 9 (computer-assisted; the unsatisfiable case is LRAT-certified)**.** *The formula $\mathcal{G}_4(4, 10, 10)$ is unsatisfiable: no four-state rule satisfying (1) synchronizes the lines of lengths $2, \ldots, 5$ in minimal time and satisfies, for each $n \in \{6, \ldots, 10\}$, (G2) and (G3) with $T = 4$, (G4) and (G5). The following variants are satisfiable: $\mathcal{G}_4(4, 30, 9)$; $\mathcal{G}_4(3, 30, 12)$; and the formula with (G1) for $T = 4$ and (G5) for $6 \leq n \leq 12$, without (G2) and (G3).*

Proof. As for Proposition 8, with the specification `full:2-5 early:6-10:4 cone:6-10`: 874 variables, 20,461 clauses, SHA-256 digest of the CNF file

`124f095450ee725be06f07271e1d752c4cbacce9f4bff190ddea1907d39e4f51`;

CaDiCaL refutes it in 36 s with an LRAT proof of 173 MB, which `lrat-check` verifies in 4 s. The variants are `full:2-5 early:6-30:4 cone:6-9`, `full:2-4 early:5-30:3 cone:5-12` and `full:2-5 cone:6-12`, for which CaDiCaL returns models. □

So with the right ends of the lines $6, \dots, 10$, four states are refuted by the complete lines $2, \dots, 5$ only, and neither the early rows nor the right ends suffice alone.

Proposition 10 (computational)**.** *Each of the following five-state rules synchronizes exactly the lines of lengths* $2, \dots, N$ *among* $2 \le n \le 60$*, for the stated* N*. The line of the stated length* $n \le 19$*, the smallest one with this property, has a cell in state* F *at the stated relative time* $\tau = t-(n-1) \le 9$*, and* $\tau < n-1$*; the last column gives the relative time of the first firing cell of every failing line up to length* 60*. The rules* δ_{13}*,* δ_{12}^{A} *and* $\delta_{13}^{\mathrm{ch}}$ *are the files* ***delta13.txt****,* ***uniformA_2-12.txt*** *and* ***delta13_chaotic.txt*** *of* ***results/****, the others are those of Proposition 5.*

rule	N	n	τ	*relative time of the first* F *in the line of length* n*,* $N < n \le 60$
δ_{13}	13	14	8	$\tau = 8$ *for every* n
δ_{12}^{A}	12	16	7	$\tau = 11, 10, 10$ *for* $n = 13, 14, 15$*;* $\tau = 7$ *for* $n \ge 16$
$\delta_{13}^{\mathrm{ch}}$	13	14	9	$7 \le \tau \le 15$*; period* 8 *in* n *for* $n \ge 22$
δ_{14}	14	15	8	$7 \le \tau \le 10$*; period* 8 *in* n *for* $n \ge 18$
δ_{14}^{b}	14	19	8	$8 \le \tau \le 14$*;* $\tau = 8, 12, 14$ *with period* 3 *for* $n \ge 25$
δ_{14}^{c}	14	15	7	$7 \le \tau \le 18$
δ_{14}^{d}	14	15	8	$6 \le \tau \le 12$
δ_{14}^{e}	14	16	9	$9 \le \tau \le 14$*;* $\tau = 9, 11$ *with period* 2 *for* $n \ge 20$

Consequently none of them satisfies $\mathcal{G}_5(9, N_1)$ *for* $N_1 \ge 19$*.*

Proof. Direct simulation of the lines of lengths $2 \le n \le 60$ (script `firstF.py`, recorded in `LOG.md`, §25). For the last claim: each rule synchronizes the lines $2, \dots, 10$, so it satisfies (G1) for $T = 9$, and the stated line violates (G2). □

For a fixed germ g, the conditions (G2) and (G3) can be imposed on all lengths $n \ge T + 2$ at once: by Lemma 1 and Corollary 3, the lengths produced by `asymp.py` represent all of them. The following models of such formulas, for the germ of δ_{14}^{d}, show how the five-state rules evade them. The rule files are in `fssp5/research/models`.

Proposition 11 (computational)**.** *Let* g *be the germ of* δ_{14}^{d}*.*

(i) The rule ***germ14d_T10*** *contains* g *and synchronizes the lines of lengths* $2, \dots, 11$ *in minimal time. No line of length* $n \ge 12$ *has a cell in state* F *at a relative time* $\tau \le 10$*. For* $12 \le n \le 16$*, no cell of the line of length* n *is* F *before time* $2n-2$*, and the cells that are* F *at time* $2n-2$ *are exactly the cells* 1 *and* 2*, and for* $n = 15$ *also the cell* 10*.*

(ii) *The rule* `d14d_T7c` *contains* g *and synchronizes the lines of lengths* $2, \dots, 8$ *and* 10*. No line of length* $n \geq 9$ *has a cell in state* F *at a relative time* $\tau \leq 7$*, and (G5) holds for* $9 \leq n \leq 20$*. For* $n \in \{9, 11, 12, \dots, 17, 19\}$*, the cells* 1 *and* n *of the line of length* n *are* F *at time* $2n-2$*, no cell is* F *before, and some cell is not* F *at time* $2n-2$*.*

(iii) *The joint formula of Lemma 15 for* $T = 11$ *is satisfiable. That formula keeps the lines* $3, \dots, 12$ *complete and the early rows (*$\tau \leq 11$*) of all lengths* $n \geq 13$*, with the entries* $(*, \mathsf{G}, \mathsf{L})$*,* $(\mathsf{G}, \mathsf{L}, \mathsf{L})$ *free. Its model* `joint15_T11` *behaves as follows. Start the line of length* $n \geq 3$ *from* $\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2}$ *at time* 1*, its configuration at time* 1 *under both labellings, and evolve it by this table. Then the lines of lengths* $3, \dots, 12$ *are synchronized at time* $2n-2$*, and no line of length* $n \geq 13$ *has a cell in state* F *at a relative time* $\tau \leq 11$*. The lines of lengths* 13 *and* 14 *have cells in state* F *at time* $2n-2$*, but not all of them. The first cell in state* F *of the lines of lengths* $15, \dots, 25$ *occurs at a relative time between* 12 *and* 17*. The table maps* $(*, \mathsf{G}, \mathsf{L})$ *and* $(\mathsf{G}, \mathsf{L}, \mathsf{L})$*, which occur again inside the reflected triangles, to* F *and* G*. So it contains neither labelling of the germ: the joint formula is a common relaxation of the two labellings, and its models need not belong to either.*

Proof. The statements about finitely many lengths are checked by direct simulation; for (ii), (G5) is checked by computing Ψ_n on the cone. The statements about all lengths $n \geq T+2$ ($T = 10, 7, 11$) follow from Lemma 1 and Corollary 3. The certificates of `asymp.py` for g are: the depth rows $0, \dots, 2T+4$ are periodic from $t_0 = 57$ with period $p = 24$ for $T = 10$, from $t_0 = 30$ with $p = 24$ for $T = 7$, and from $t_0 = 57$ with $p = 48$ for $T = 11$. The half-line of g is the same under both labellings and does not contain the neighbourhoods $(*, \mathsf{G}, \mathsf{L})$, $(\mathsf{G}, \mathsf{L}, \mathsf{L})$ at times $t \geq 1$ (Lemma 15), so in (iii) it is the half-line used by the joint formula. Moreover, the representative lengths and all lengths up to 200 were simulated and have no F at $\tau \leq T$. The rules were found as models of the formulas `full:2-11` plus the early rows ($T = 10$) of the representatives; `full:2-8` plus the early rows ($T = 7$) and `cone:9-20`; and the joint formula of Lemma 15 at $T = 11$ (`full:3-12` plus the early rows). □

So the early rows control the right part of each line at small relative times, and the right-end cones control the right end at the firing time. Both are satisfied by rules whose lines fire only at their two ends, as in (ii). To exclude such rules, a formula must make middle cells fire, and the dependency cone of a middle cell of the firing row covers most of R_n (for $n = 15$, 76 of the 120 cells).

3 Uniform pre-firing configurations

Lemma 12. *Let* $X \in W$*,* $N \geq 3$*, and let* δ *be a minimal-time solution such that for every* $3 \leq n \leq N$ *the configuration of the line of length* n *at time* $2n-3$ *is* X^n*. Define* δ' *by* $\delta'(x, y, z) := \mathsf{L}$ *for every* $(x, y, z) \in W^3 \setminus \{(X, X, X)\}$ *with* $\delta(x, y, z) = \mathsf{F}$*, and* $\delta' := \delta$ *elsewhere. Then* $\delta'(X, X, X) = \mathsf{F}$*, no other neighbourhood without a border is mapped to* F *by* δ'*, and* δ' *coincides with* δ *on every neighbourhood that occurs in the half-line or in a line of length* $n \leq N$ *before time* $2n-2$*. In particular,* δ' *satisfies every constraint that the lines of lengths* $2, \dots, N$ *and the half-line impose on* δ*.*

Proof. The rule δ' satisfies the quiescence condition (1) of the note, since $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$. The line of length 3 fires from XXX, so $\delta(X, X, X) = \mathsf{F} = \delta'(X, X, X)$. Let $(x, y, z) \in W^3 \setminus \{(X, X, X)\}$ with $\delta(x, y, z) = \mathsf{F}$. It does not occur in the half-line, which never fires (equation (2) of the note).

If it occurred as the neighbourhood of a cell of the line of length $n \le N$ at a time $t < 2n-2$, that cell would be F at time $t+1$; this is impossible for $t+1 < 2n-2$, and for $t = 2n-3$ the configuration is X^n, whose neighbourhoods without a border are all (X, X, X) (the line of length 2 has none). Changing the values of neighbourhoods that never occur changes neither the half-line nor these lines. □

So if the constraints of the lengths $2, \dots, N$ (with the half-line encoded to any anti-diagonal), together with "the only neighbourhood without a border that gives F is (X, X, X)", are unsatisfiable, then no minimal-time solution fires from X^n at every length $3 \le n \le N$. This is option `-Pint XXX` of `gcegar.py`; option `-upf X` adds the implied unit clauses "every cell at time $2n-3$ is X". The state $X = \mathsf{L}$ is impossible, because $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$, and $X = \mathsf{B}$ is the case $X = \mathsf{A}$ after exchanging the auxiliary states. Mazoyer's solution fires from G^n, and so do the rules satisfying Balzer's condition (B3) in the strong reading.

4 A pumping lemma for the right band

The left-border lemma (Lemma 4.1 of the note) runs a band of chains along an anti-diagonal. The same pigeonhole argument applies to the cells next to the right border, read in the direction of time. It gives a necessary condition on the reflected triangle itself.

Lemma 13 (right band)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $1 \le d \le \tau_0 \le n-2$ and $\ell \ge 1$, and suppose that the column $\tau \mapsto R_n(\tau, d)$ is ℓ-periodic on $[\tau_0, n-2]$. Then $n - 1 - \tau_0 \le (k-1)^d \ell$.*

Proof. For $d-1 \le \tau \le n-1$ put $\mathbf{b}(\tau) := (R_n(\tau, 0), \dots, R_n(\tau, d-1))$. Since $R_n(\tau+1, \kappa) = \delta(R_n(\tau, \kappa+1), R_n(\tau, \kappa), R_n(\tau, \kappa-1))$ with $R_n(\tau, -1) = *$, there is a map Γ, depending on δ and d only, with $\mathbf{b}(\tau+1) = \Gamma(\mathbf{b}(\tau), R_n(\tau, d))$ for $d \le \tau \le n-2$; its arguments lie in $W^d \times W$ because no cell fires before time $2n-2$. At $\tau = n-1$ the band is F^d. Suppose $n - 1 - \tau_0 > (k-1)^d \ell$. The pairs $(\mathbf{b}(\tau), \tau \bmod \ell)$, $\tau_0 \le \tau \le n-2$, take at most $(k-1)^d \ell$ values, so there are $\tau_0 \le \tau_1 < \tau_2 \le n-2$ with $\mathbf{b}(\tau_1) = \mathbf{b}(\tau_2)$ and $\ell \mid h := \tau_2 - \tau_1$. For $0 \le s \le n-2-\tau_2$ the driving cells $R_n(\tau_1 + s, d)$ and $R_n(\tau_2 + s, d)$ lie in $[\tau_0, n-2]$ and differ by a multiple of the period, so they are equal, and induction on s gives $\mathbf{b}(\tau_1 + s) = \mathbf{b}(\tau_2 + s)$ for $0 \le s \le n-1-\tau_2$. For $s = n-1-\tau_2$ this reads $\mathbf{b}(n-1-h) = \mathbf{b}(n-1) = \mathsf{F}^d$, although $n-1-h < n-1$: a contradiction. □

Unlike the two barriers of the note, the hypothesis concerns cells of R_n, not of the half-line, so the lemma is no test on germs. The partial-triangle tests (cones of firing cells, `gfix.py`) imply it for every length they encode.

5 The architecture of Mazoyer's solution

The following facts about Mazoyer's six-state solution (file `results/mazoyer6.txt`, extracted from Duprat's Coq development) are established by simulation; they motivate the classes of rules tested in `LOG.md`.

Proposition 14 (computational)**.** *Let δ_M be Mazoyer's rule and let U be the set of neighbourhoods that occur in the lines of lengths $3, \dots, 120$ before their firing time.*

(i) $\delta_M(*, y, z) = \delta_M(\mathsf{L}, y, z)$ for every $(*, y, z) \in U$ (five neighbourhoods), and $\delta_M(x, y, *) = \delta_M(x, y, \mathsf{G})$ for every $(x, y, *) \in U$ (fifteen neighbourhoods); for every other state X at least eight of the fifteen right-border entries differ from $\delta_M(x, y, X)$.

(ii) $\delta_M(\mathsf{L},\mathsf{L},x)=\mathsf{L}$ *for every working state* x.

(iii) Let $H(n)$ *be the largest* T *such that every cell of* R_n *with* $\kappa\le\tau-4$ *and* $\tau\le T$ *is* L*: up to the relative time* $H(n)$*, the reflected triangle is quiescent behind an echo of width four. Then* $H(n)=4,4,5,6,8,9,11,14,21,31,41,201$ *for* $n=8,10,12,15,20,25,30,40,60,90,120,600$.

So the left border of Mazoyer's solution acts as a quiescent cell and the right border as a general, and the echo of the right end clears the line behind it until it meets the division point at about one third of the line from the right end ($H(n)\approx n/3$), where a new general starts a shorter line in the cleared cells. None of the five-state rules δ_{14}, $\delta_{14}^{\mathrm{b}},\dots,\delta_{14}^{\mathrm{e}}$ has either property: each uses border transitions that differ from the interior transition with the border replaced by any state, and behind their echoes the reflected triangles never become quiescent (Proposition 5 and `LOG.md`, §§20–21).

6 The complexity-15 half-line

The two germs of complexity 15 that survive the tests of the note are those of δ_{14}^{c} and δ_{14}^{d}. Let σ be the renaming $\mathsf{A}\mapsto\mathsf{G}$, $\mathsf{B}\mapsto\mathsf{A}$, $\mathsf{G}\mapsto\mathsf{B}$ of the states of δ_{14}^{d} (L and F fixed).

Lemma 15. *σ maps the thirteen transitions of the germ of δ_{14}^{d} that occur at times $t\ge1$ onto the thirteen transitions of the germ of δ_{14}^{c} that occur at times $t\ge1$. Let $n\ge3$. In the names of δ_{14}^{d}, the line of length n is $\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2}$ at time 1 for both germs. Hence the constraints that the line of length n imposes on a completion of the germ of δ_{14}^{d} and, transported by σ^{-1}, on a completion of the germ of δ_{14}^{c} are the same. The only exceptions are the four entries $(*,\mathsf{G},\mathsf{L})$, $(\mathsf{G},\mathsf{L},\mathsf{L})$, which the germ of δ_{14}^{d} fixes to A,B, and $(*,\mathsf{A},\mathsf{L})$, $(\mathsf{A},\mathsf{L},\mathsf{L})$, which the transported germ of δ_{14}^{c} fixes to A,B.*

Proof. The germ of δ_{14}^{d} is $*$GL>A, GLL>B, $*$AB>A, ABL>B, BLL>B, ABB>B, BBL>L, BLB>B, LBL>A, BBA>L, BAB>B, LBB>L, BBB>B, LBA>B, ABA>A. That of δ_{14}^{c} is $*$GL>G, GLL>A, $*$GA>G, GAL>A, ALL>A, GAA>A, AAL>L, ALA>A, LAL>G, AAG>L, AGA>A, LAA>L, AAA>A, LAG>A, GAG>G. Applying σ to the last thirteen transitions of the first list gives the last thirteen of the second. In both germs the first two transitions are used only at $t=0$ (the general is the only cell in state G of δ_{14}^{d}'s half-line, and $(*,\mathsf{G},\mathsf{L})$, $(\mathsf{G},\mathsf{L},\mathsf{L})$ do not occur in δ_{14}^{c}'s half-line at $t\ge1$, by the correspondence). At time 1 the line of length $n\ge3$ is $(\delta(*,\mathsf{G},\mathsf{L}),\delta(\mathsf{G},\mathsf{L},\mathsf{L}),\mathsf{L},\dots,\mathsf{L})$, which is $\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2}$ for δ_{14}^{d} and $\mathsf{G}\,\mathsf{A}\,\mathsf{L}^{n-2}$, that is $\sigma(\mathsf{A}\,\mathsf{B}\,\mathsf{L}^{n-2})$, for δ_{14}^{c}. From time 1 on, the line evolves by the table alone, so a completion δ of the first germ and the completion $\sigma\circ\delta\circ\sigma^{-1}$ of the second (after the four entries above are set as stated) produce corresponding lines of every length $n\ge3$. The line of length 2 reads $(\mathsf{G},\mathsf{L},*)$ at time 0 in both germs, which corresponds to two different entries. □

Consequently the completion problems of the two labellings for the lengths $3,\dots,N$, with the four entries left free, have a common relaxation; if it is unsatisfiable, both germs are refuted and every five-state minimal-time solution has $c_{78}\ge16$.